Transactra

Every transfer balanced. Every action traced.

A Database-First Banking Simulation

Initial Database Documentation & ERDs

CSAI_202_Course_Project

Introduction to Database Systems

Zewail City of Science and Technology

School of Computational Sciences and Artificial Intelligence (CSAI)

Version 1.0

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1. System Overview and Scope

Transactra is a simulated bank whose database, not its website, guarantees that money is never lost, duplicated, or overspent. Customers apply for accounts and send simulated money; operations officers review applicants; compliance analysts investigate rule-based fraud alerts; and managers audit and reconcile the books. All data is synthetic.

The database must store six kinds of information:

Domain	What it stores	Entities
Identity and access	Who can sign in and which role they hold	AppUser, AppRole, AppUserRole
Customer onboarding	Customer profiles, applications, documents, screening	Customer, KycTier, KycApplication, KycDocument, WatchlistEntry, ScreeningResult
Accounts	Accounts of three kinds and their status history	Account, CheckingAccount, SavingsAccount, WalletAccount, AccountStatusHistory
Payments and ledger	Transfers, double-entry postings, retry protection	Transfer, LedgerEntry, IdempotencyRecord
Compliance	Fraud rules, alerts, and investigation cases	FraudRule, Alert, AlertTransfer, AlertAccount, ComplianceCase, CaseAlert, CaseNote, CaseAction
Governance	Audit trail, integrity checks, settings	AuditLog, ReconciliationRun, ReconciliationException, SystemSetting

Out of scope for the database: real money, real identity documents (only metadata is stored), real watchlists, and connections to outside banks or payment networks.

2. Real-World Model

The ER diagram follows one continuous story; each step below names the entities it introduces.

1. **Registration.** Mona creates a login (**AppUser**) and receives the Customer role (**AppRole** through **AppUserRole**). She fills in her profile (**Customer**).
2. **Application.** She submits an onboarding request (**KycApplication**) listing identity documents by metadata only (**KycDocument**).
3. **Screening and approval.** Officer Karim compares her name against a fictional list (**WatchlistEntry**); each run is stored (**ScreeningResult**). He approves her and assigns a risk tier (**KycTier**) that sets her transfer limits.

4. **Accounts.** Karim opens accounts for her (**Account**), each exactly one of **CheckingAccount**, **SavingsAccount**, or **WalletAccount**. Every status change is logged (**AccountStatusHistory**).
5. **Money moves.** Her salary arrives and she pays her friend Omar. Each payment is a **Transfer** from a source account to a destination account, posting one debit and one matching credit (**LedgerEntry**). A request key (**IdempotencyRecord**) ensures a retried request never pays twice. A mistaken transfer is undone by a second, linked reversal transfer.
6. **Fraud monitoring.** A detection rule (**FraudRule**) notices five accounts paying one beneficiary within minutes and raises an **Alert**, linked to the transfers (**AlertTransfer**) and accounts (**AlertAccount**) involved.
7. **Investigation.** Analyst Laila groups related alerts into a **ComplianceCase** (**CaseAlert**), writes **CaseNote** entries, and records each step (**CaseAction**) until she resolves it.
8. **Oversight.** Manager Hana reviews every important action (**AuditLog**), runs an integrity check (**ReconciliationRun**) that lists any problems (**ReconciliationException**), and adjusts configuration (**SystemSetting**).

3. Entities

The model has 22 strong entities (including the Account supertype), 4 associative (junction) entities, and 3 subtypes. Primary keys are marked PK in the diagrams; full attribute definitions are in the Scenarios and Entity Dictionary document.

	Entity	Kind	Description	Primary key	Main attributes
1	AppUser	Strong	A login identity for a customer or staff member	UserId	Username, Email, PasswordHash, Status, CreatedAtUtc
2	AppRole	Strong	A permission group: Customer, Operations, Compliance, Admin	RoleId	RoleName, Description
3	AppUserRole	Associative	Assignment of a role to a user	UserId + RoleId	AssignedByUserId, AssignedAtUtc
4	Customer	Strong	A bank client's profile	CustomerId	FullName, DateOfBirth, Phone, Address, CountryCode, RiskLevel
5	KycTier	Strong	A risk tier and its transfer limits	TierId	TierName, PerTransferLimit, DailyLimit, IsActive
6	KycApplication	Strong	One onboarding request and its decision	ApplicationId	SubmissionNo, Status, SubmittedAtUtc, DecisionReason, DecidedAtUtc

	Entity	Kind	Description	Primary key	Main attributes
7	KycDocument	Strong	Metadata of one identity document	DocumentId	DocType, Status, DocumentRef, IssueDate, ExpiryDate
8	WatchlistEntry	Strong	One fictional name on the screening list	EntryId	FullName, NormalizedName, SourceLabel, IsActive
9	ScreeningResult	Strong	One screening run of an application	ResultId	RunAtUtc, Result, MatchScore, MatchReason, ReviewStatus
10	Account	Strong (supertype)	Any place money is held	AccountId	AccountNumber, AccountType, Status, CurrencyCode, IsInternal, MaintainedBalance, OpenedAtUtc
11	CheckingAccount	Subtype of Account	Spending account with overdraft rules	AccountId	OverdraftEnabled, OverdraftLimit
12	SavingsAccount	Subtype of Account	Savings account with a minimum balance	AccountId	InterestRateLabel, MinimumBalance
13	WalletAccount	Subtype of Account	Lightweight payment account	AccountId	WalletAlias, MaxBalance
14	AccountStatusHistory	Strong	One status change of an account	HistoryId	OldStatus, NewStatus, Reason, ChangedAtUtc
15	Transfer	Strong	One money-movement request and its outcome	TransferId	TransferRef, Amount, CurrencyCode, Kind, Status, ReasonCode, Memo, RequestedAtUtc, CompletedAtUtc
16	LedgerEntry	Strong	One permanent debit or credit posting	EntryId	EntryType, Amount, CurrencyCode, PostedAtUtc
17	IdempotencyRecord	Strong	A request key that blocks duplicate execution	Idempotency Id	IdemKey, RequestFingerprint, Status, CreatedAtUtc
18	FraudRule	Strong	One transparent detection rule	RuleId	RuleCode, Name, WindowMinutes, CountThreshold, AmountThreshold, Version

	Entity	Kind	Description	Primary key	Main attributes
19	Alert	Strong	One firing of a rule	AlertId	RuleVersion, Severity, Status, Summary, DedupeKey, CreatedAtUtc
20	AlertTransfer	Associative	Links an alert to a triggering transfer	AlertId + TransferId	(none)
21	AlertAccount	Associative	Links an alert to an involved account	AlertId + AccountId	InvolvedAs
22	ComplianceCase	Strong	An investigation grouping alerts	CaseId	Title, Status, OpenedAtUtc, ClosedAtUtc, Resolution
23	CaseAlert	Associative	Links an alert to a case	CaseId + AlertId	LinkedByUserId, LinkedAtUtc
24	CaseNote	Strong	One append-only investigation note	NoteId	NoteText, CreatedAtUtc
25	CaseAction	Strong	One recorded step on a case	ActionId	ActionType, FromStatus, ToStatus, Detail, CreatedAtUtc
26	AuditLog	Strong	One important action, append-only	AuditId	Action, EntityType, EntityId, Outcome, Metadata, OccurredAtUtc
27	ReconciliationRun	Strong	One integrity check	RunId	StartedAtUtc, FinishedAtUtc, Status, ExceptionCount
28	ReconciliationException	Strong	One problem found by a run	ExceptionId	ExceptionType, ExpectedValue, ActualValue, Detail
29	SystemSetting	Strong	One configuration value	SettingKey	SettingValue, Description

Derived attribute: an account's balance can be derived from its ledger entries (credits minus debits). MaintainedBalance stores it for speed and is checked by reconciliation.

4. Relationships

Each relationship is named with a verb read from the first entity to the second. Cardinality is (first side):(second side); participation states whether the second entity must take part (total) or may not (partial).

4.1 Business Relationships

#	Relationship	First entity	Second entity	Cardinality	Participation of second
R1	holds	AppUser	AppRole	M:N (via AppUserRole)	Partial
R2	logs in as	AppUser	Customer	1:0..1	Partial
R3	is classified as	Customer	KycTier	N:1	Partial (set on approval)
R4	submits	Customer	KycApplication	1:N	Total
R5	includes	KycApplication	KycDocument	1:N	Total
R6	is screened in	KycApplication	ScreeningResult	1:N	Total
R7	is matched in	WatchlistEntry	ScreeningResult	1:N	Partial
R8	owns	Customer	Account	1:N	Partial (internal accounts have no owner)
R9	is a	Account	Checking / Savings / Wallet	Specialization, disjoint and total	Total
R10	changes status in	Account	AccountStatusHistory	1:N	Total
R11	sends	Account (source)	Transfer	1:N	Total
R12	receives	Account (destination)	Transfer	1:N	Total
R13	reverses	Transfer	Transfer	1:0..1 (recursive)	Partial
R14	posts	Transfer	LedgerEntry	1:0..2	Total

#	Relationship	First entity	Second entity	Cardinality	Participation of second
R15	is affected by	Account	LedgerEntry	1:N	Total
R16	is protected by	Transfer	IdempotencyRecord	1:0..1	Partial
R17	raises	FraudRule	Alert	1:N	Total
R18	cites	Alert	Transfer	M:N (via AlertTransfer)	Partial
R19	involves	Alert	Account	M:N (via AlertAccount)	Partial
R20	groups	ComplianceCase	Alert	M:N (via CaseAlert)	Partial
R21	has notes	ComplianceCase	CaseNote	1:N	Total
R22	records	ComplianceCase	CaseAction	1:N	Total
R23	finds	ReconciliationRun	ReconciliationException	1:N	Total
R24	flags account	Account	ReconciliationException	1:N	Partial
R25	flags transfer	Transfer	ReconciliationException	1:N	Partial

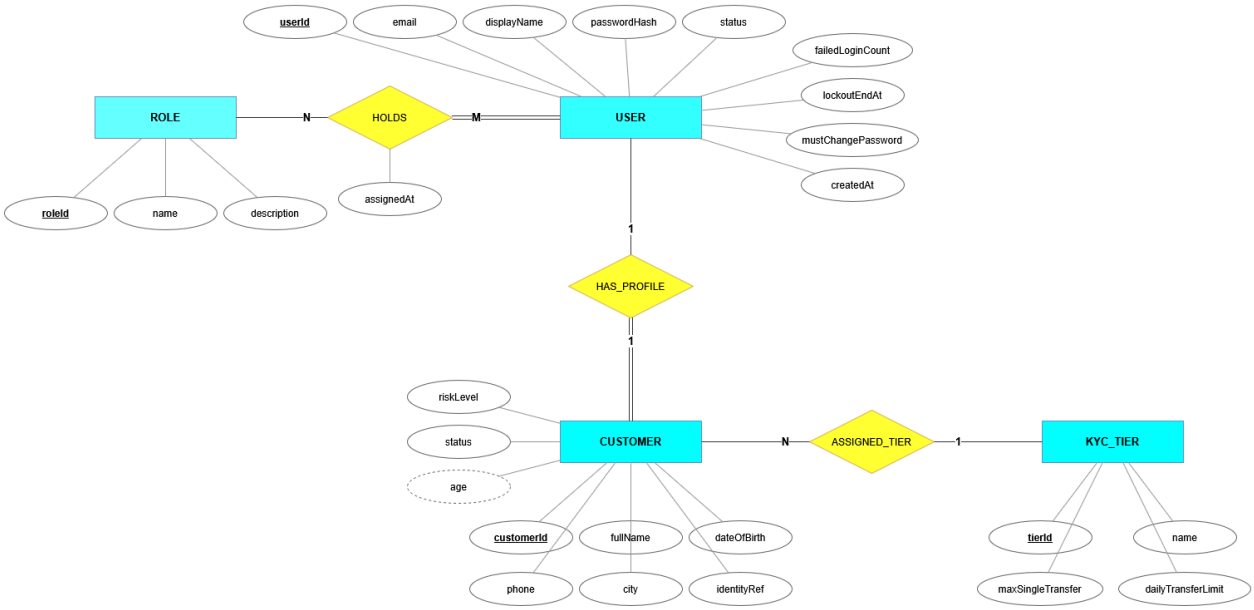
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5. ER Diagrams

The full model is shown as one overview and four domain diagrams, because 29 entities in a single drawing would be unreadable. Shared entities (Account, Transfer) appear in more than one diagram to show the links between domains. **(All of the following ERDs were made via Draw.io)**

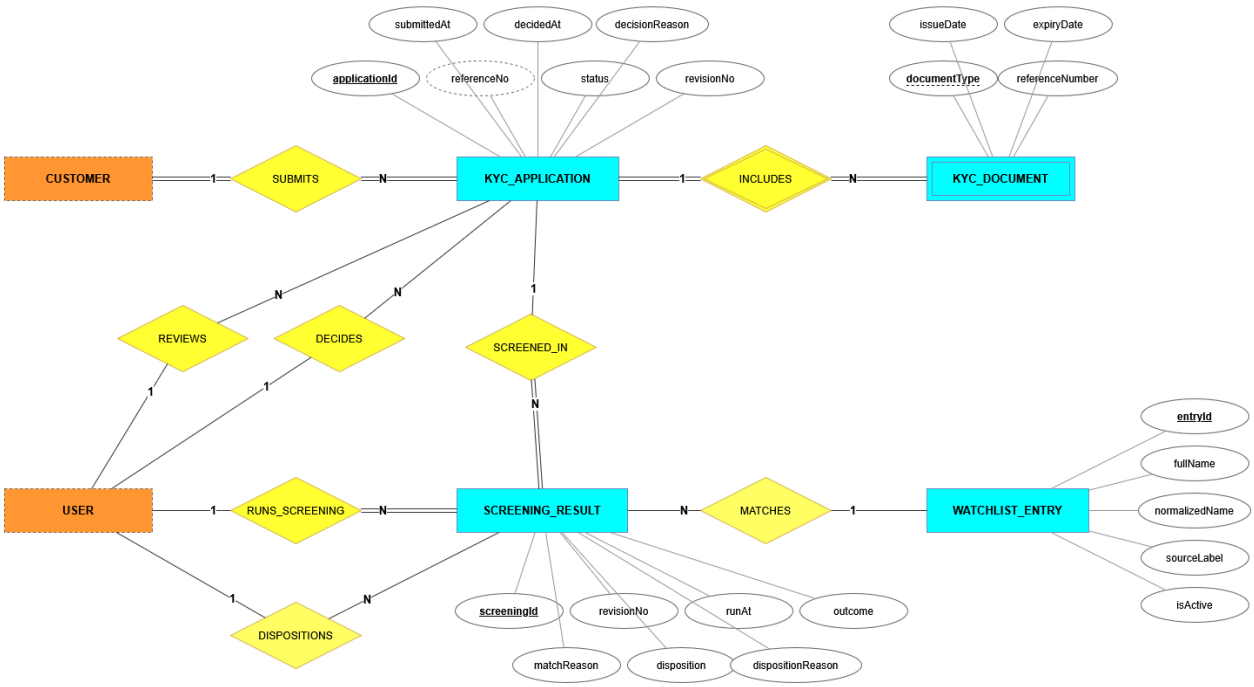
5.1 ERD 1: Users, Roles and Customers:

ER-1 Users, Roles and Customers



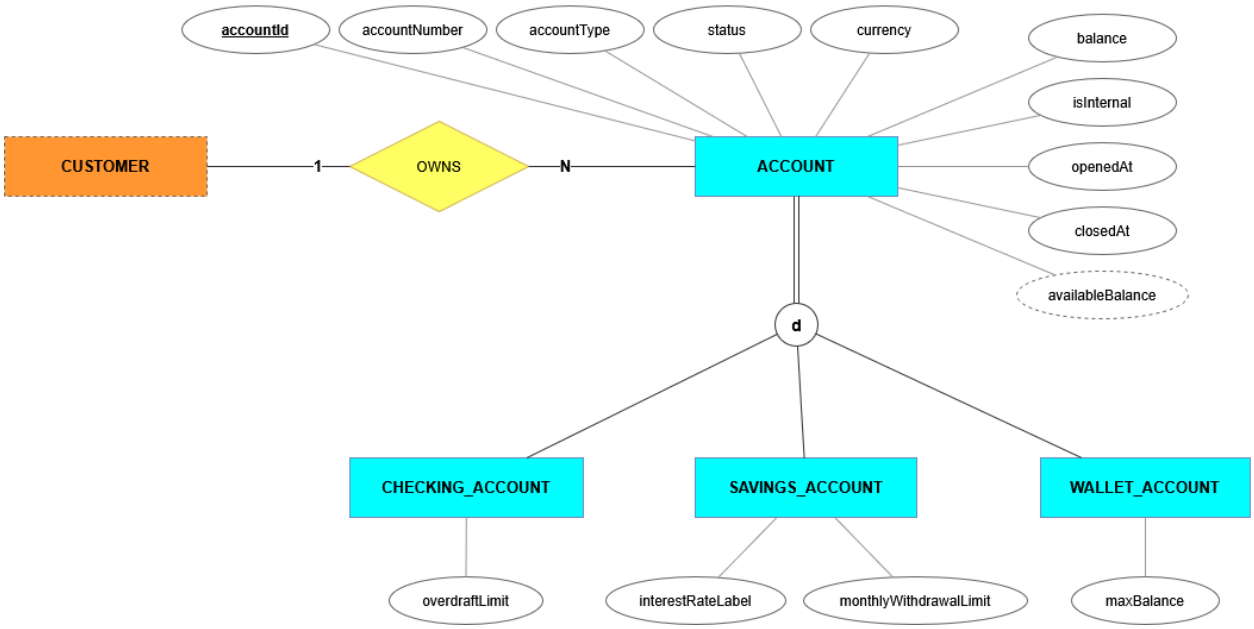
5.2 ERD 2: Customer Onboarding:

ER-2 Customer Onboarding (KYC)



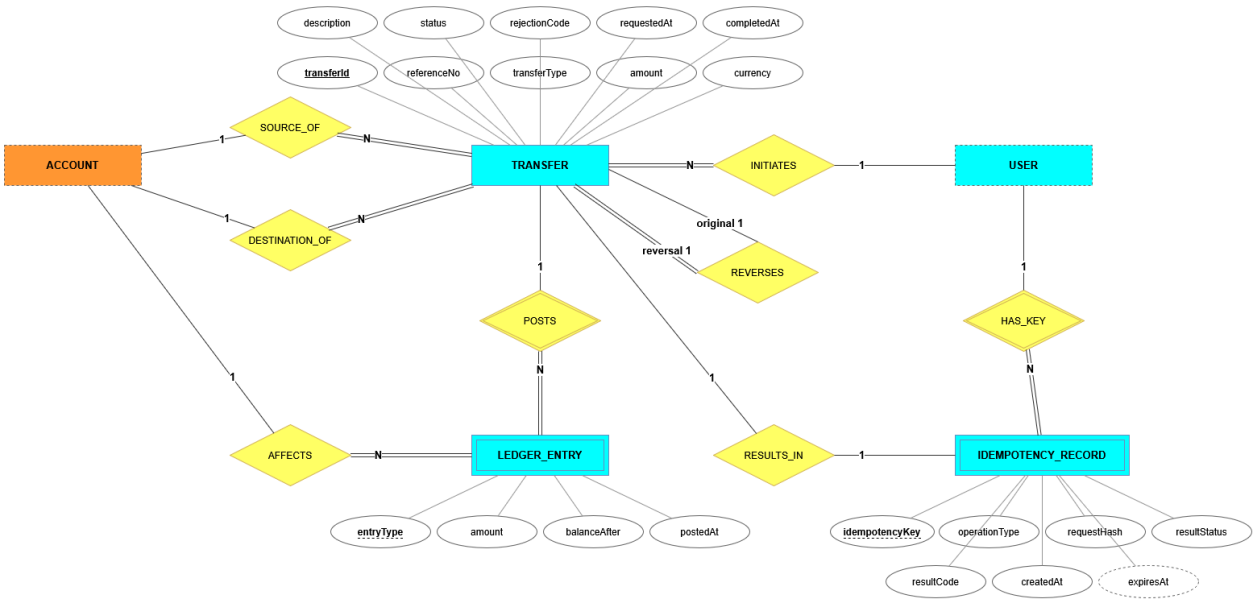
5.3 ERD 3: Accounts and Specialization

ER-3 Accounts and Specialization



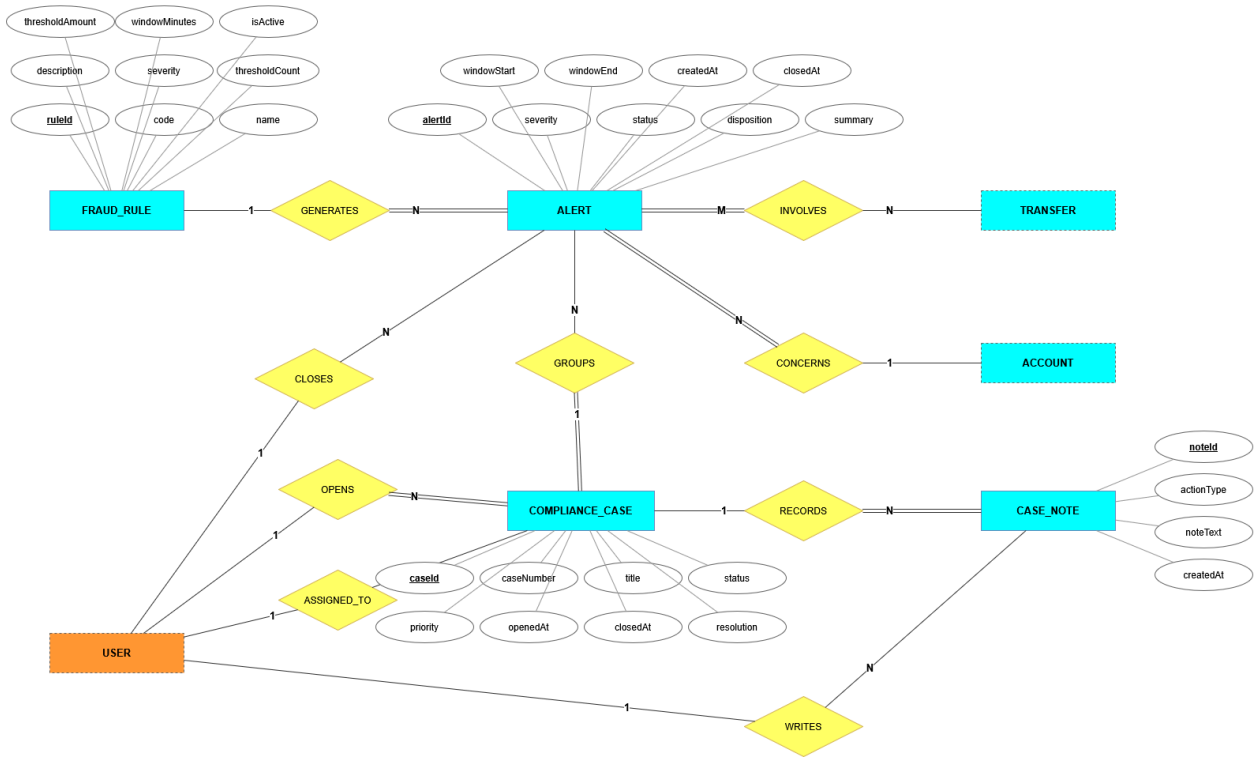
5.4 ERD 4: Transfer & Ledger:

ER-4 Transfers, Ledger and Idempotency



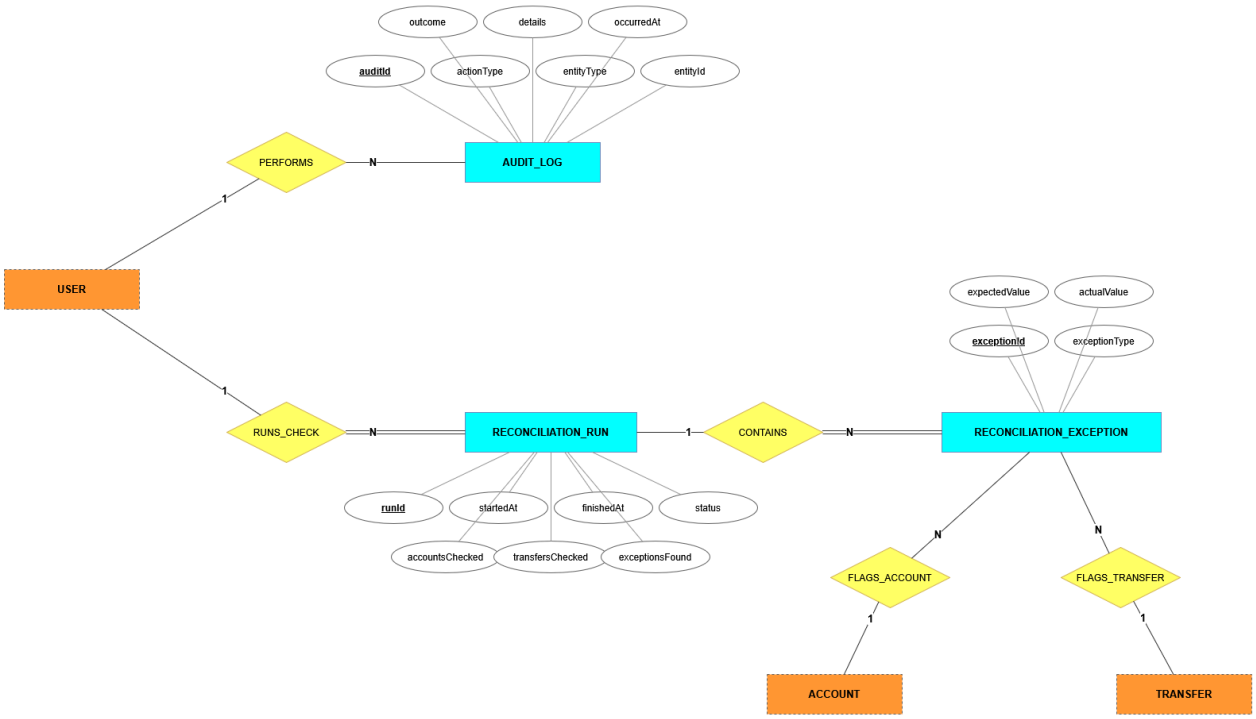
5.5 ERD 5: Fraud Monitoring & Compliance:

ER-5 Fraud Monitoring and Compliance



5.6 ERD 6: Audit & Reconciliation:

ER-6 Audit and Reconciliation



Orange dashed entities: attributes in ER-1, ER-3 and ER-4.